Multimodality @ Hugging Face

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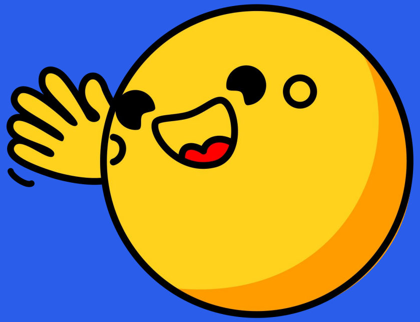
Machine Learning Engineer @ Hugging Face

Content

- What's HF?
- What's multimodality?
- Why is it interesting now?
- How do we choose the best?
- How do we build/use with it?



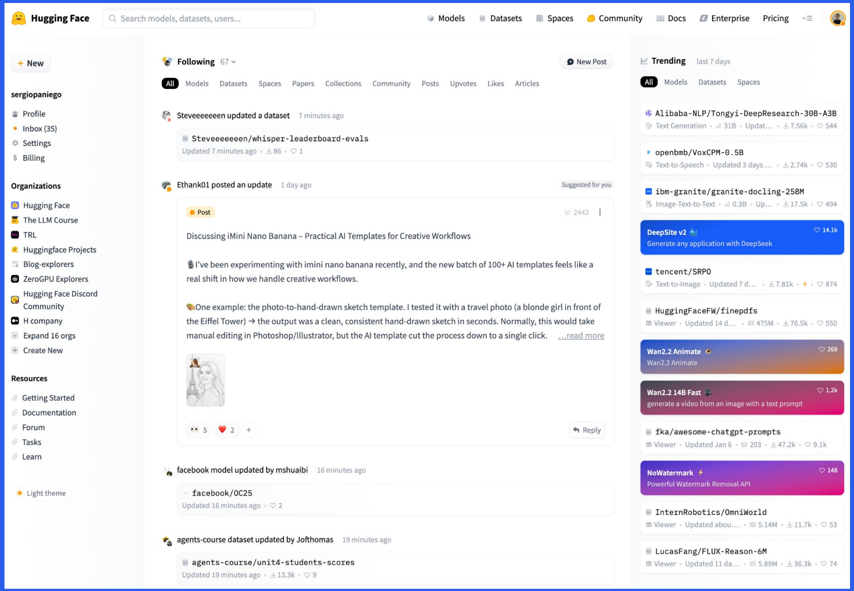
Hi, I'm Sergio 👋

- Machine Learning Engineer @ Hugging Face
    - Multimodality, Post-Training & Agents
- PhD in AI
- Open source and AI → GSoC
- Teaching and Sharing

What's Hugging Face? 🤗

AI open platform (+community) where AI developers share (HF Hub):

- Models, datasets, apps…
- Posts, blogs, courses, papers, leaderboards…



2.1M
public models

500K+
public datasets

1M+
daily downloads

1M+
daily visitors

What's Hugging Face? 🤗

Set of open source projects to empower AI development and advancement

Datasets

Transformers

Trackio

Hub

LeRobot

TRL

30+

Libraries

Diffusers

Transformers.js

Accelerate

timm

PEFT

smolagents

Open vs Closed 🔓

	Open	Closed / Proprietary
Security	Models can be **self-hosted**, data stays in your environment	Models cannot be self-hosted. Data is **sent outside** your environment to vendor
Control	The timing and nature of updates are **controlled** by you	Updates and changes to performance can happen **without notice**
Customization	**Full source code access** to customize the model for your needs	**Limited ability to customize** for your needs
Transparency	**Inspect code and data** provides better auditability and understandability	**No ability to audit** or understand performance
Cost	Typical **lower long term cost** due to smaller model size	larger model size and **proprietary premium** often balanced by **decreased** cost from **server-side optimization**
Latency	**Lower latency** due to on premise and smaller model sizes	Often **greater latency** due to larger model sizes + API latency
Quality	No single approach is best. Each use case will vary. Proprietary is typically **closer to the frontier of performance**.	
Examples	OpenAI, Meta, deepseek, MISTRAL AI_, Microsoft	OpenAI, ANTHROPIC

How HF democratizes ML/AI 🤝

ML engineers who train models from scratch

Software engineers who use inference APIs, ML engineers who fine-tune models

Anyone who can use a GUI

🤗 Datasets

🤗 Models

🤗 Spaces

Used everywhere in the AI world

15,000+ startups and enterprises

grammarly, HUAWEI, Dropbox, INTUIT, Spotify, T, orange, databricks, Apple, servicenow

Meta, Microsoft, ORACLE, NVIDIA, salesforce, Bloomberg, HubSpot, Klarna., Uber, SAMSUNG

Open source contributors

Meta, MISTRAL AI_, stability.ai, Microsoft, Ai2, OpenAI

Hugging Face

Cloud partners

aws, Azure, NVIDIA, CLOUDFLARE

Hardware partners

NVIDIA, AMD, aws, intel, habana, GRAPHCORE, Qualcomm

On-prem partners

DELL, IBM

What's multimodality?

Previously:

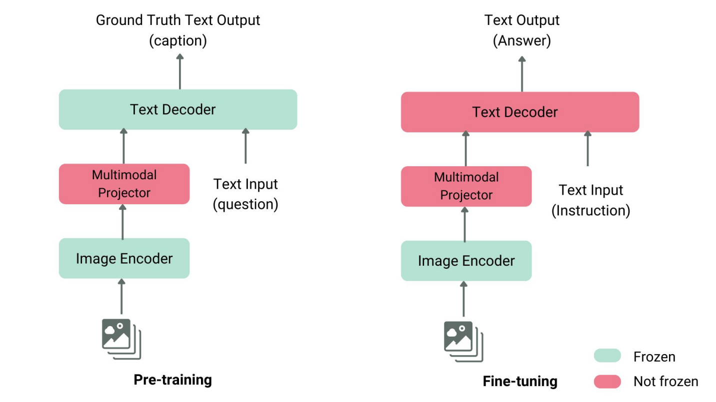
- Text-to-text 📝 → 📝

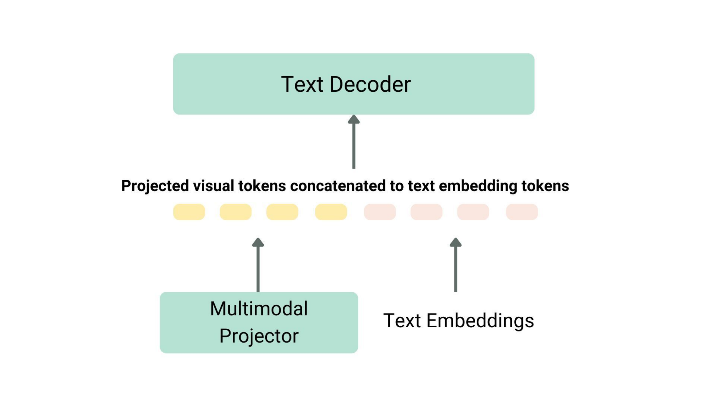
Combining 2 or more modalities:

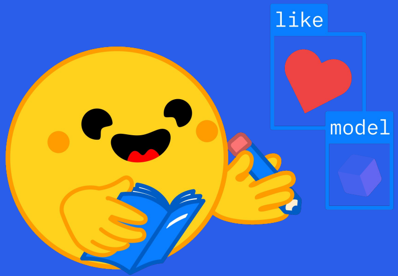
- Text 📝
- Image 🖼️
- Audio 🔊
- Video 🎥

Vision Language Models (images+text)

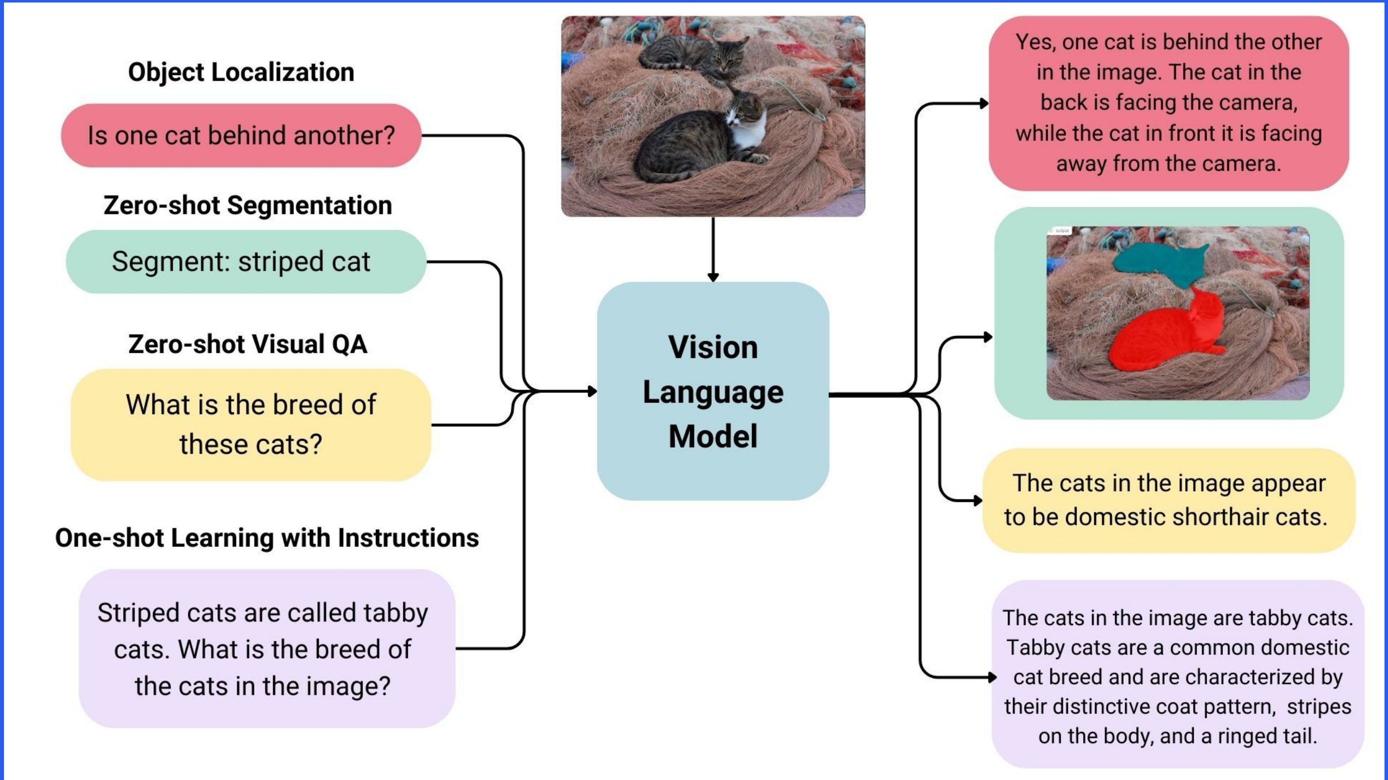
Omni Models (any combination)



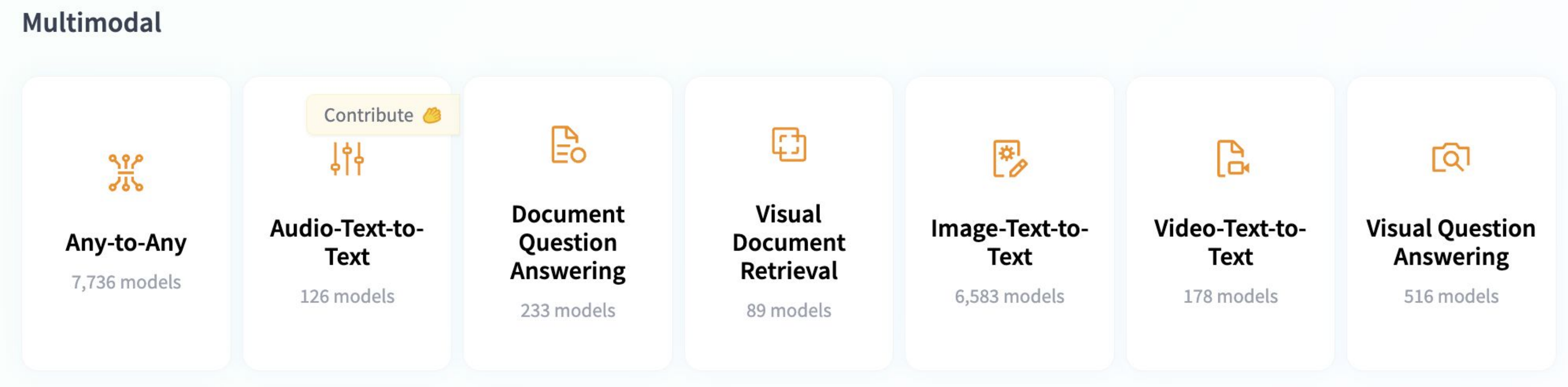




Vision Language Models

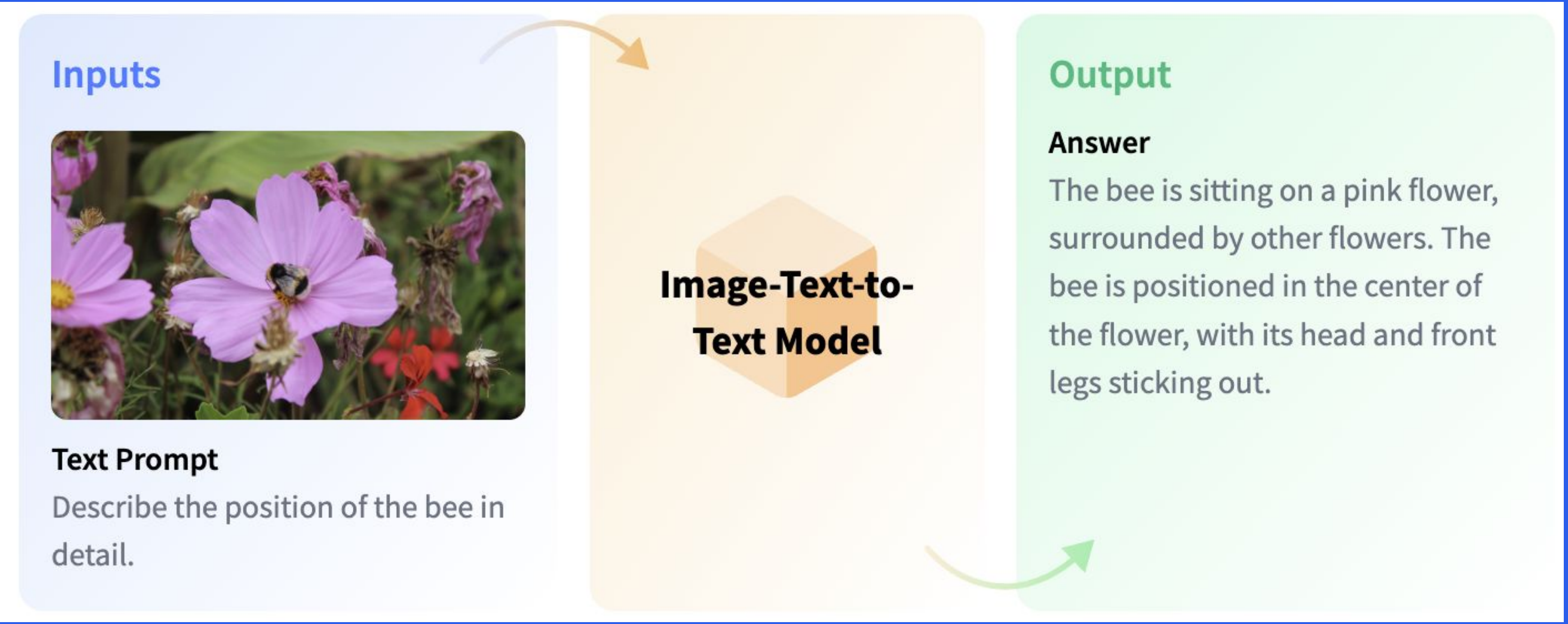


Multimodal tasks in Hugging Face



hf.co/tasks

Image-text-to-text (VLMs)



https://huggingface.co/tasks/image-text-to-text

Image-text-to-text (VLMs)

- Qwen3-VL/Qwen2.5-VL
- SmolVLM2
- Gemma 3n
- Pixtral
- GLM-4.5V
- Moondream 3
- Llama 4
- InternVL3.5
- Kimi-VL-A3B
- MiniCPM-V-4_5
- DeepSeek-VL2

HF collection

Vision Language Models: 2025 Update

updated May 12

This collection includes all the models, datasets and Spaces mentioned in the blog Vision Language Models: 2025 Update

Qwen/Qwen2.5-Omni-7B
Any-to-Any • 11B • Updated Apr 30 • 176k • 1.79k

Qwen2.5 Omni 7B Demo 🏆 — 349
Generate text and speech from audio, video, and text inputs

Qwen2.5-Omni Technical Report
Paper • 2503.20215 • Published Mar 26 • 165

openbmb/MiniCPM-o-2_6
Any-to-Any • 9B • Updated Jun 20 • 155k • 1.25k

deepseek-ai/Janus-Pro-7B
Any-to-Any • Updated Feb 1 • 190k • 3.5k

Chat With Janus-Pro-7B — 2k
A unified multimodal understanding and generation model.

Janus-Pro: Unified Multimodal Understanding and Generation with Data and Model Scaling
Paper • 2501.17811 • Published Jan 29 • 6

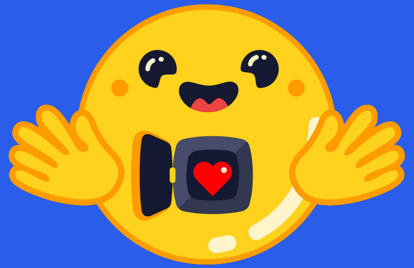
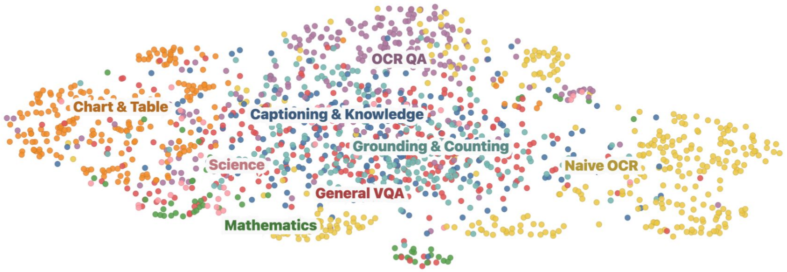


Image-text-to-text (VLMs)

https://huggingface.co/spaces/sergiopaniego/vlm_object_understanding

Not only VLM models!

FineVision:
Open Data Is All You Need



A new open dataset for data-centric training of Vision Language Models

AUTHORS
Luis Wiedmann
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Thibaud Frere
Leandro von Werra

AFFILIATION
Hugging Face

PUBLISHED
Sep 4, 2025

PDF
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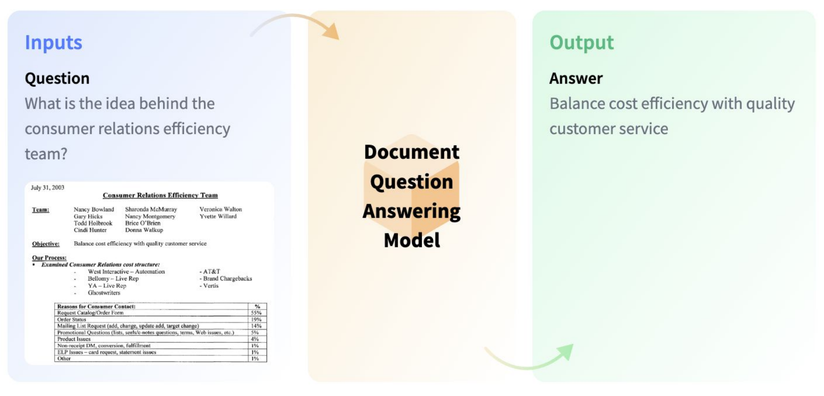
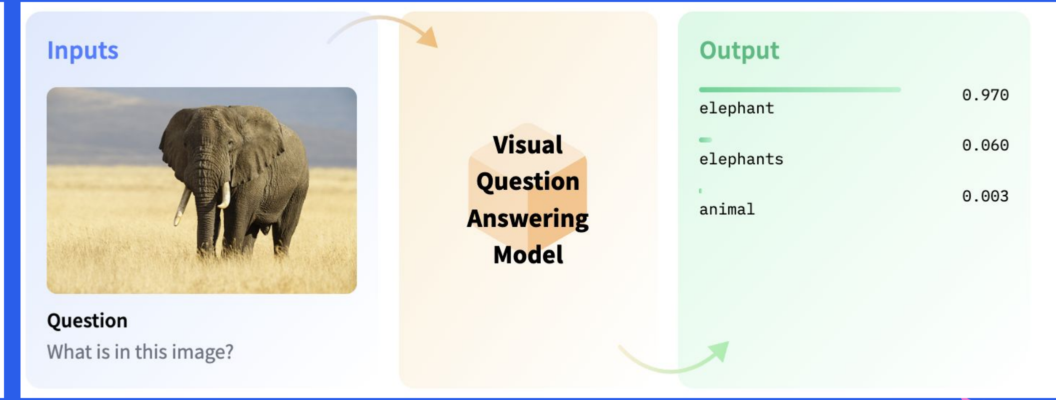
huggingface.co/datasets/HuggingFaceM4/FineVision

Today, we release **FineVision**, a new multimodal dataset with **24 million samples**. We created FineVision by collecting over **200 datasets** containing **17M images**, **89M question-answer turns**, and **10B answer tokens**, totaling **5TB of high-quality data**. Additionally, we extensively processed all datasets to unify their format, clean them of duplicates and poor data, and rated all turns using 32B VLMs across 4 qualitative metrics with a score from 1-5 to enable the construction and study of individual training mixtures.

https://huggingface.co/spaces/HuggingFaceM4/FineVision

Document/Visual QA

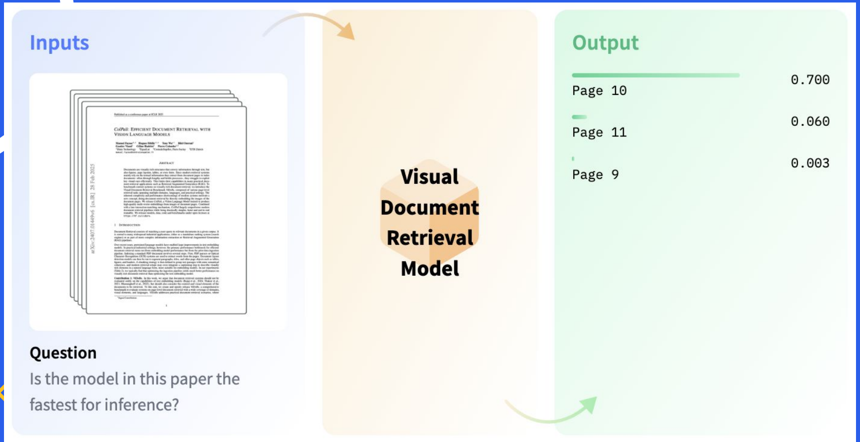


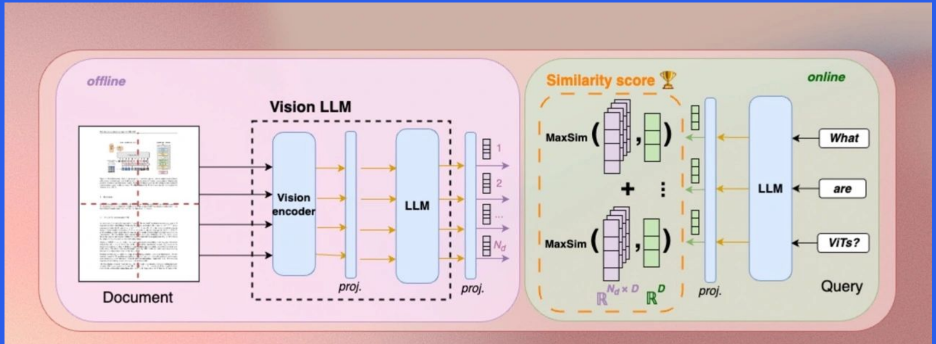


https://huggingface.co/tasks/document-question-answering
https://huggingface.co/tasks/visual-question-answering

Visual Document Retrieval (multimodal RAG)

ColPali, ColQwen2...





ColPali's Architecture

https://huggingface.co/tasks/visual-document-retrieval

Any-to-Any (Omni)

Inputs

Text Prompt

What is the significance of this place?

Any-to-Any Model

Output

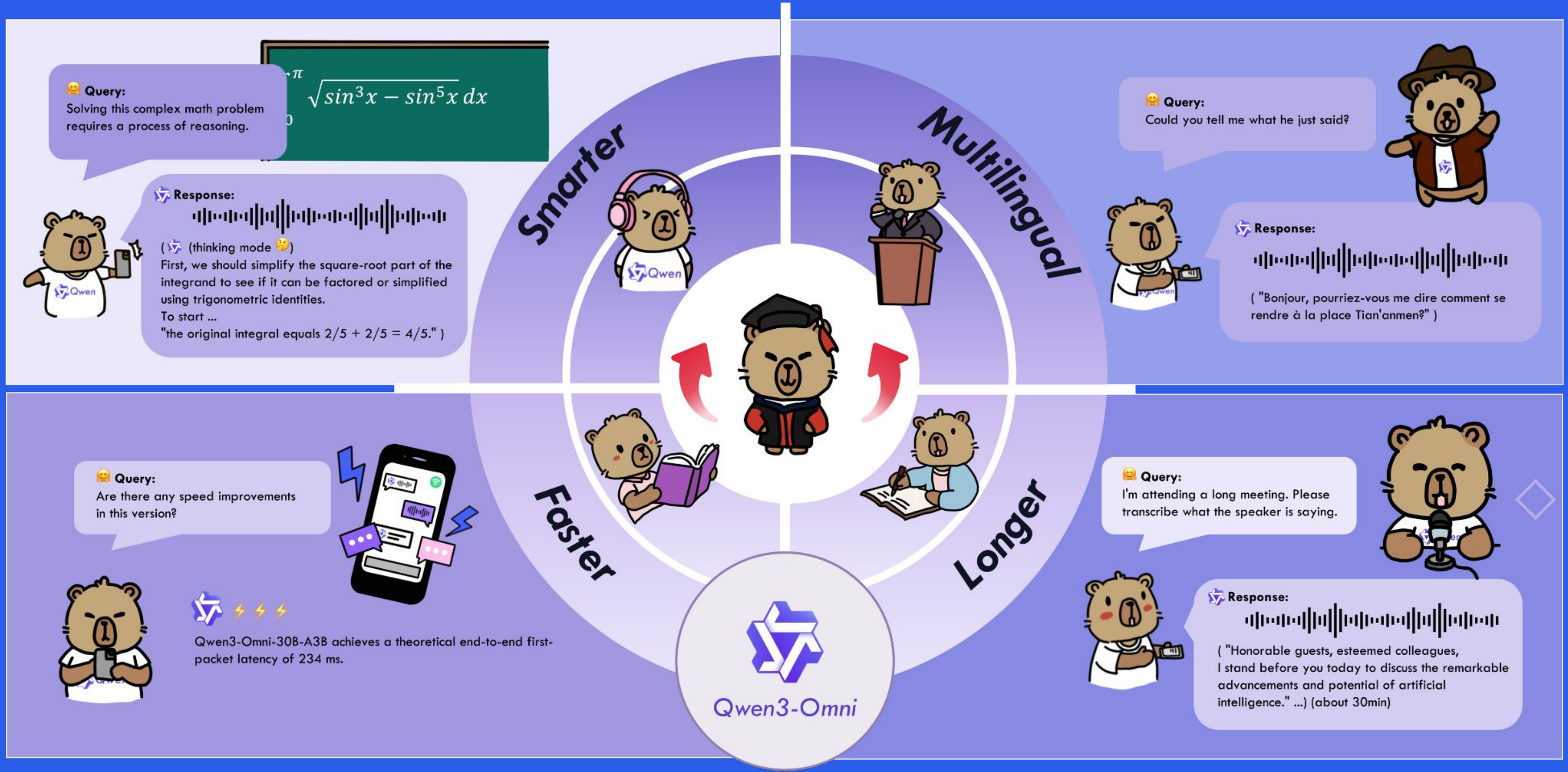
Generated Text

The place in the picture is Osaka Castle, located in Osaka, Japan. Osaka Castle is a historic castle that was originally built in the 16th century by Toyotomi Hideyoshi, a powerful warlord of the time. It is one of the most famous landmarks in Osaka and is known for its distinctive white walls and black roof tiles. The castle has been rebuilt several times over the centuries and is now a popular tourist attraction, offering visitors a glimpse into Japan's rich history and culture.

0:00 / 0:29

https://huggingface.co/tasks/any-to-any

Any-to-Any (Omni)

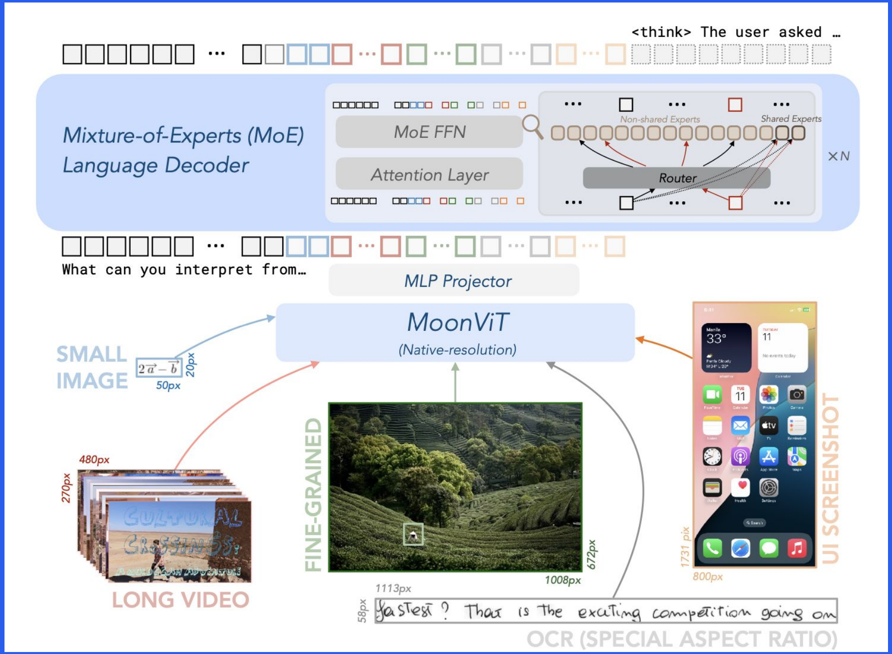


Any-to-Any (Omni)

https://huggingface.co/spaces/Qwen/Qwen3-Omni-Demo

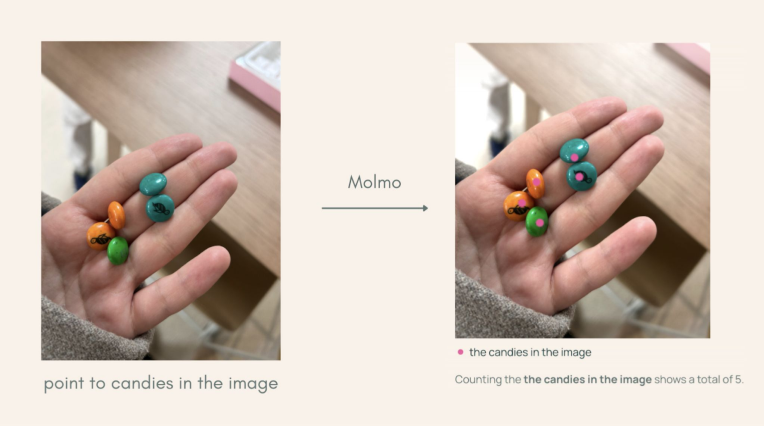
Trends

- **Reasoning models:**
  - QVQ-72B-preview
  - Kimi-VL-A3B-Thinking
- **Smol models**
  - Gemma 3n
  - smolVLM2
  - Qwen2.5-VL-3B-Instruct
  - Moondream 3B
  - PerceptronAI/Issac-0.1
- **MoEs**



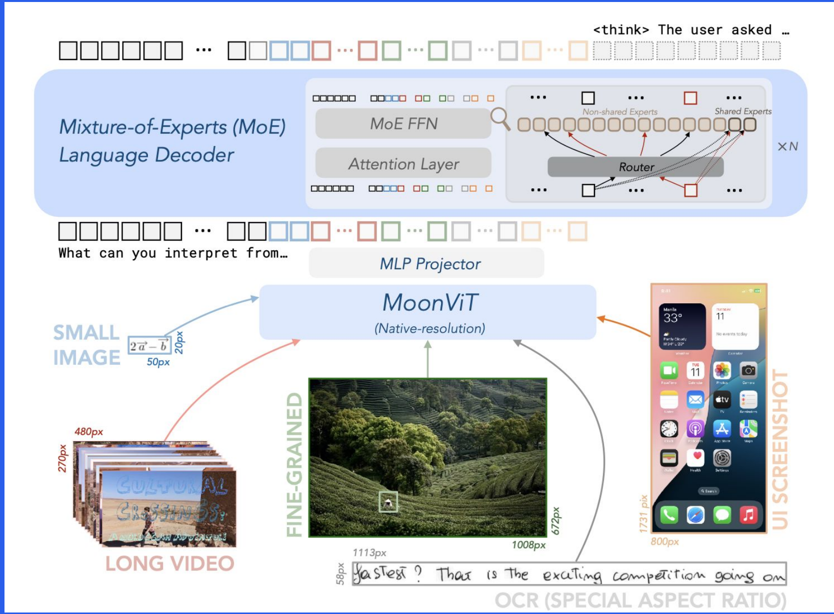
Trends

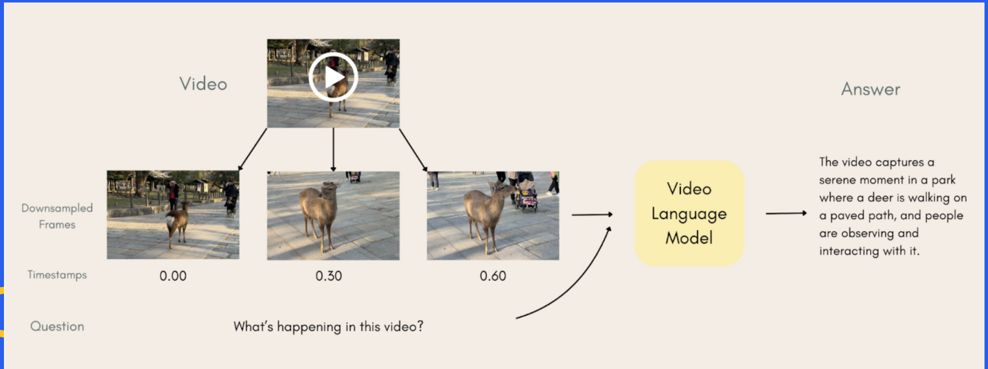
- **Vision-Language-Action (VLAs)**
    - SmolVLA, GR00T N1, ShowUI, SimLingo
- **Specialized models** (Object Detection, Segmentation, Counting)
    - PaliGemma, Molmo, moondream3, Issac0.1



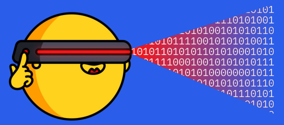
Trends

- **Agentic tasks:**
  - Kimi-VL-A3B-Thinking
  - Qwen2.5-VL-32B
- **Video Language Models:**
  - SmolVLM2, LongVU, Qwen2.5VL, Gemma3n





Choosing the best



- **General:** MMMU-Pro, MMT-Bench…
- **Specialized benchmarks**
- **OpenVLM Leaderboard**
- **Trade-offs:** size 📏, speed ⚡, quality ⭐, licensing 🔒

Model Name

Input the Model Name (fuzzy, case insensitive)

Model Size

☐ <4B ☐ 4B-10B ☐ 10B-20B ☐ 20B-40B ☑ >40B ☐ Unknown

Model Type

☑ API ☑ OpenSource

Rank	Method	Param (B)	Language Model	Vision Model	Eval Date	Avg Score	Avg Rank	MMBench_V11	MMStar	MMMU_VAL	MathVista	OCRBench	AI2D	HallusionBench	MMVet
1	InternVL3-78B	78.4	Qwen2.5-72B	InternViT-6B-v2.5	2025/04/14	79.1	5.88	87.7	73.4	72.2	79	908	89.8	59.1	80.7
2	InternVL2.5-78B-MPO	78	Qwen-2.5-72B	InternViT-6B-v2.5	2024/12/29	77	12.88	87.7	72.1	68.2	76.6	909	89.2	58.1	73.5
3	Qwen2.5-VL-72B	73.4	Qwen2.5-72B	QwenViT	2025/02/02	76.1	19	87.8	70.5	68.2	74.2	882	88.5	54.6	76.9
4	BailingMM-Pro-0120	81	Bailing-80B	SigLIP-400M	2025/01/25	75.2	21.62	87.1	69.4	59.2	72.3	894	87.2	59.4	77.6
5	Qwen2-VL-72B	73.4	Qwen2-72B	QwenViT	2024/10/28	74.8	24.38	85.9	68.6	64.3	69.7	888	88.3	58.7	73.9
6	Qwen-VL-Max-0809	72	Qwen2-72B	QwenViT	2024/09/12	74.4	26.5	85.8	69.2	64.6	68.3	881	88.1	59.2	72.3
7	InternVL2-Llama3-76B	76	Llama-3-70B-Instruct	InternViT-6B	2024/07/19	71	46.25	85.5	67.1	58.3	65.6	842	87.6	55.4	64.4
8	InternVL2-40B	40	Nous-Hermes-2-Yi-34B	InternViT-6B	2024/07/11	69.7	54	85	64.7	55.2	64.3	833	86.8	56.5	61.8
9	InternVL2.5-78B	78	Qwen-2.5-72B	InternViT-6B-v2.5	2024/12/10	68.3	51.75	87.5	69.5	70	70.6	853	89.1	57.4	16.9

https://huggingface.co/spaces/opencompass/open_vlm_leaderboard

Choosing the best (Vibes)

https://huggingface.co/spaces/visionLMsftw/comparevlms

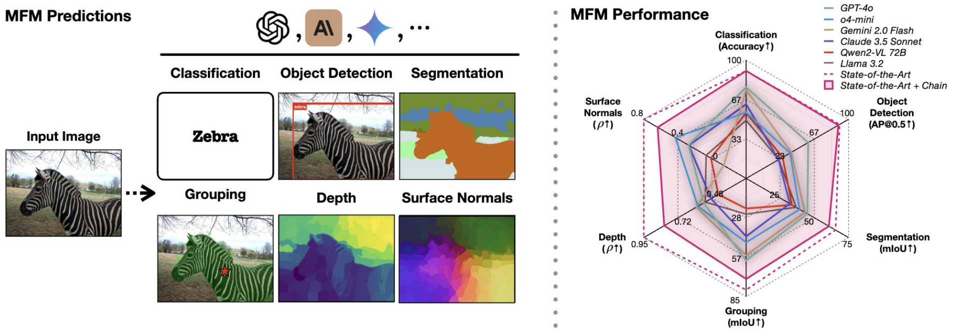
Everything done by a single

How Well Does GPT-4o Understand Vision? Evaluating Multimodal Foundation Models on Standard Computer Vision Tasks

Rahul Ramachandran, Ali Garjani, Roman Bachmann,
Andrei Atanov *, Oğuzhan Fatih Kar *, Amir Zamir *
Swiss Federal Institute of Technology (EPFL)

* Equal technical advising.

Paper | Code | Slides | Twitter

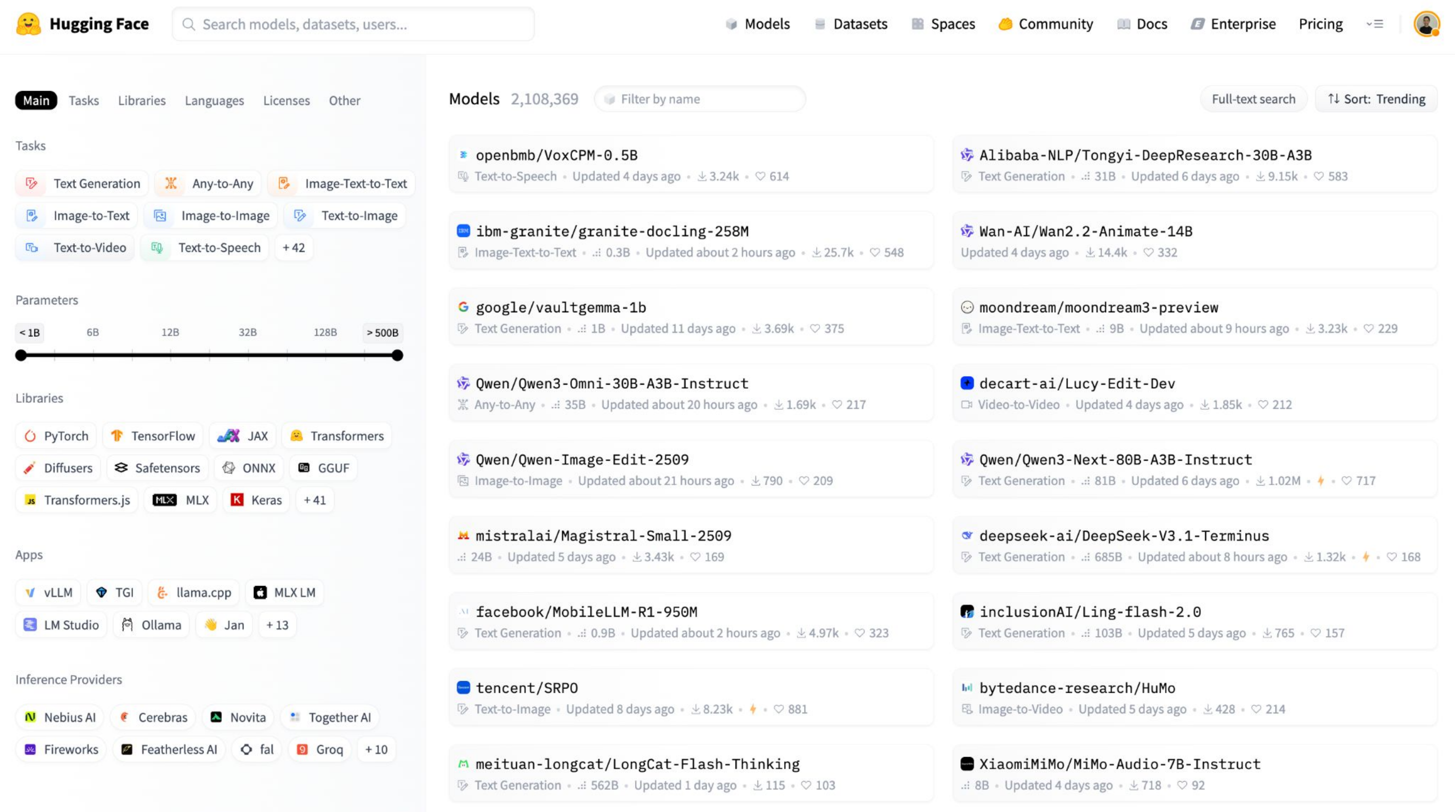


https://huggingface.co/papers/2507.01955

How do we build with it?

Find model → Customize → Deploy

Find the model



https://huggingface.co/models

Use it with 🤗 Transformers

The Transformers Library: standardizing model definitions

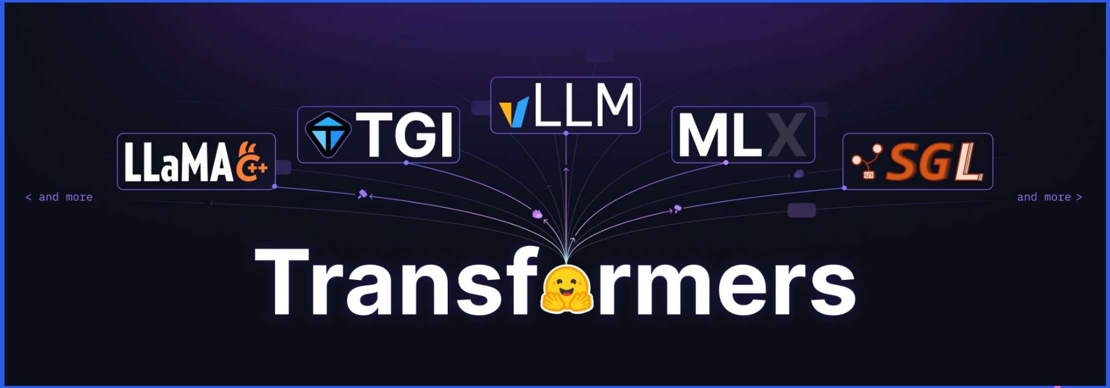
Published May 15, 2025

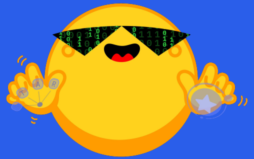
Update on GitHub

Lysandre lysandre Follow · Arthur Zucker ArthurZ Following · Pedro Cuenca pcuenq Following · Julien Chaumond julien-c Following

TLDR: Going forward, we're aiming for Transformers to be the pivot across frameworks: if a model architecture is supported by transformers, you can expect it to be supported in the rest of the ecosystem.

Transformers was created in 2019, shortly following the release of the BERT Transformer model. Since then, we've continuously aimed to add state-of-the-art architectures, initially focused on NLP, then growing to Audio and computer vision. Today, transformers is the default library for LLMs and VLMs in the Python ecosystem.





https://huggingface.co/blog/transformers-model-definition

Use it with 🤗 Transformers

```python
from transformers import pipeline

pipe = pipeline("image-text-to-text")
messages = [
    {
        "role": "user",
        "content": [
            {
                "type": "image",
                "image": "https://huggingface.co/datasets/huggingface/documentation-images/resolve/main/bee.jpg",
            },
            {"type": "text", "text": "Describe this image."},
        ],
    }
 ]
outputs = pipe(text=messages, max_new_tokens=60, return_full_text=False)
print(outputs[0]["generated_text"])
# The image depicts a close-up view of a vibrant and colorful flower. The flower in focus is a large, pink, five-petal
flower with a prominent yellow center, which is characteristic of many species of the genus Dianthus. The petals are
broad and slightly curved, giving the flower a full
```

https://huggingface.co/docs/transformers/index

Use it with 🤗 Transformers

```python
#pip install qwen-vl-utils[decord]==0.0.8
from transformers import AutoProcessor, AutoModelForImageTextToText
from qwen_vl_utils import process_vision_info
import torch

model_name = "Qwen/Qwen2.5-VL-3B-Instruct"
processor = AutoProcessor.from_pretrained(model_name)
model = AutoModelForImageTextToText.from_pretrained(model_name, torch_dtype="auto", device_map="auto")

messages = [
    {
        "role": "user",
        "content": [
            {"type": "image", "image": "https://huggingface.co/datasets/huggingface/documentation-images/resolve/main/bee.jpg"},
            {"type": "text", "text": "Describe this image."},
        ],
    }
]

text = processor.apply_chat_template(
    messages, tokenize=False, add_generation_prompt=True
)
image_inputs, _ = process_vision_info(messages)
inputs = processor(
    text=[text],
    images=image_inputs,
    padding=True,
    return_tensors="pt",
)
inputs = inputs.to("cuda")

generated_ids = model.generate(**inputs, max_new_tokens=128)
generated_ids_trimmed = [out_ids[len(in_ids) :] for in_ids, out_ids in zip(inputs.input_ids, generated_ids)]
output_text = processor.batch_decode(generated_ids_trimmed, skip_special_tokens=True, clean_up_tokenization_spaces=False)
print(output_text[0])
# The image depicts a close-up view of several flowers in a garden setting. The flowers are predominantly pink with some having
yellow centers, and they appear to be cosmos or similar types of daisy-like plants. The petals are delicate and slightly ruffled,
giving them a soft appearance. Some of the flowers have fully bloomed, while others are still in bud form.
```

Customize with TRL

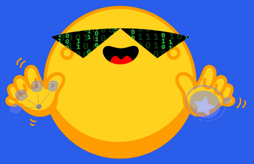
TRL allows you to fine-tune VLMs using different methods: SFT, DPO, MPO, GRPO, GSPO, RLOO, and Online DPO.
PEFT and quantization support

```
from trl import SFTConfig, SFTTrainer
from datasets import load_dataset

trainer = SFTTrainer(
    model="Qwen/Qwen2.5-VL-3B-Instruct",
    args=SFTConfig(max_length=None),
    train_dataset=load_dataset("trl-lib/llava-instruct-mix", split="train"),
)
trainer.train()
```

https://huggingface.co/docs/trl/index

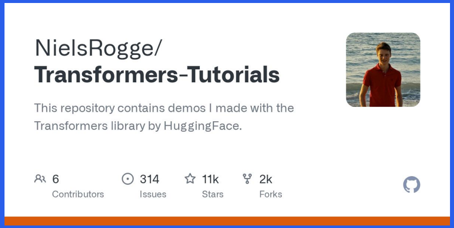
Deploy model

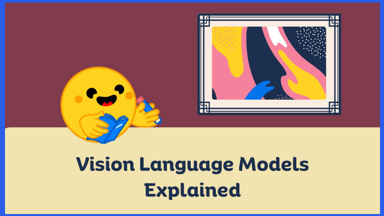


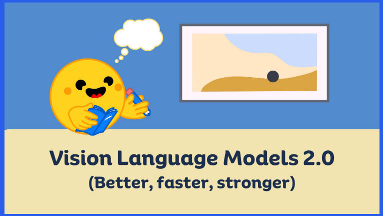
Inference and serving frameworks
Inference Providers (TGI)
Hugging Face Endpoints

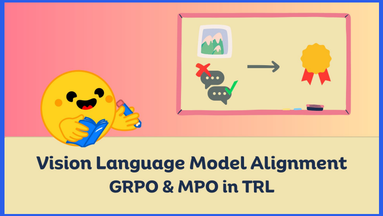
Resources

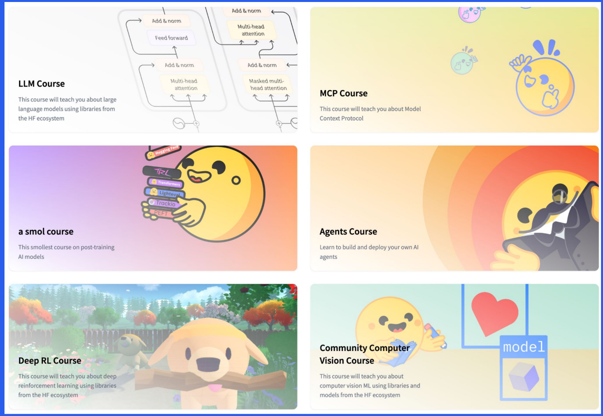
- hf.co/blog
- hf.co/learn
- Transformers-Tutorials repo
- smol-vision repo











Closing

- Hugging Face: hub, libraries, and community for AI
- Multimodal AI: combining text, images, audio, & video
- Benchmarks & evaluation
- Tools for use, customization, and deployment: Transformers, TRL, Spaces...

Thanks!

Sergio Paniego Blanco (@sergiopaniego)

Machine Learning Engineer @ Hugging Face